Foundations for Cryptographic Reductions in CCSA Logics

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Context

Security protocols
critical components
of communication systems

Security properties

on protocols:
Authentication, Secrecy,...

on primitives:
CCA, EUF-MAC,...

Formal verification

$$H : \mathsf{P} \models \psi$$

assuming H, protocol P verifies property ψ

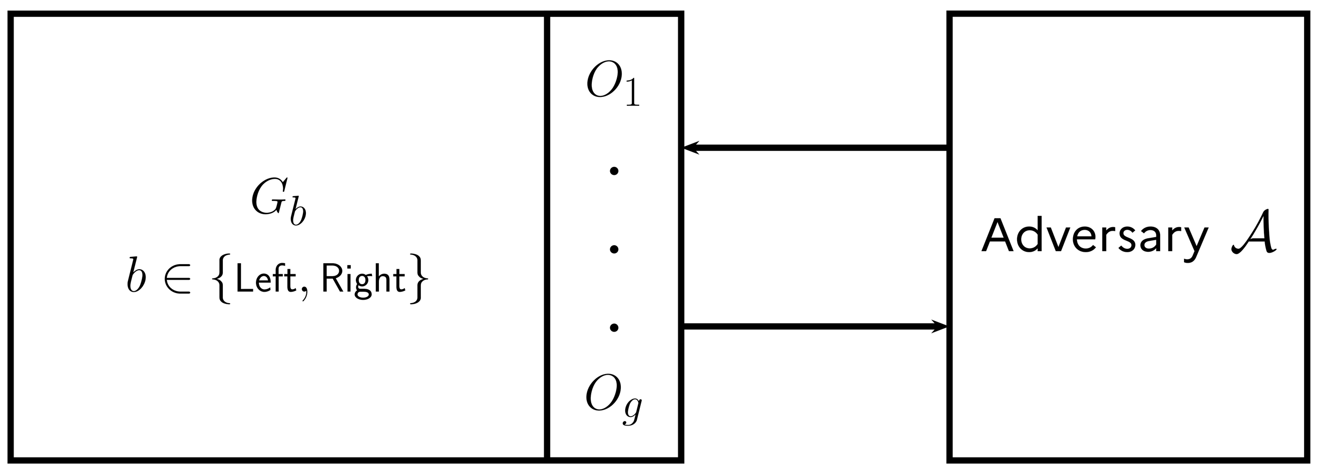
Mechanization

Computer aided and verified proofs

$\rightarrow$ Strengthen confidence in results

Cryptographic reduction

Cryptographic game



Game indistinguishability

$|\Pr(\mathcal{A}(G_{\text{Left}}) = 1) - \Pr(\mathcal{A}(G_{\text{Right}}) = 1)|$ is negligible

Cryptographic reduction

For all $\mathcal{A}$, there exists a poly-time simulator $\mathcal{S}$ s.t.

$\mathcal{A}$ breaks protocol $\Rightarrow$ $\mathcal{S}(\mathcal{A})$ breaks primitive

Bideduction Predicate

Constraint system
register randomness usage

Bi-terms
input, output

$$\mathcal{C}, \{\psi, \varphi\} \vdash \vec{u} \rhd_{\mathsf{G}} \vec{v}$$

Pre- and post-conditions
on G's memory

Semantics

There exists s such that whenever c is consistent:

- Randomness is used according to c.
- From any memory in φ, executing s yields a memory in ψ.
- $s(\vec{u}_{\text{Left}}) = \vec{v}_{\text{Left}}$ and $s(\vec{u}_{\text{Right}}) = \vec{v}_{\text{Right}}$.

Proof System for Bideduction

Function application

$$\frac{\mathcal{C}, \{\varphi, \psi\} \vdash \vec{u} \rhd \vec{v} \qquad \vdash \mathsf{adv}(g)}{\mathcal{C}, \{\varphi, \psi\} \vdash \vec{u} \rhd g(\vec{v})}$$

$$\mathcal{S}() := \mathsf{x}_v \leftarrow \mathcal{S}_1(); \quad \mathsf{x}_{res} \leftarrow g(\mathsf{x}_v)$$

Sampling

$$\frac{}{\mathcal{C} \cdot (s, \mathsf{T}_S), \{\varphi, \psi\} \vdash \vec{u} \rhd s}$$

$$\mathcal{S}() := \mathsf{x}_{res} \xleftarrow{\$}$$

Oracle call

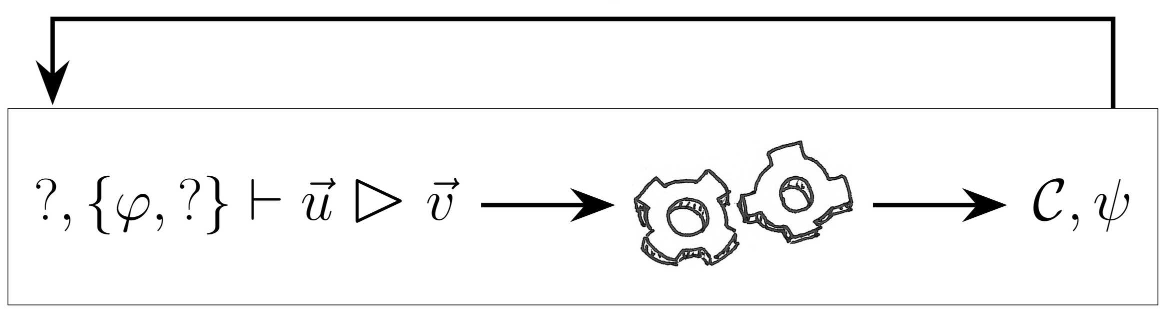
$$\frac{\mathcal{C}, \{\varphi, \varphi'\} \vdash \vec{u} \rhd \vec{v} \qquad \{\varphi'\}\mathsf{O}_i(\vec{v})[\vec{s}, \vec{k}]\{\psi\}}{\mathcal{C}', \{\varphi, \psi\} \vdash \vec{u} \rhd \mathsf{O}_i(\vec{v})[\vec{s}, \vec{k}]}$$

with $\mathcal{C}' = \mathcal{C} \cdot (\vec{s}, \mathsf{T}^{\text{loc}}_{\mathsf{G}}) \cdot (\vec{k}, \mathsf{T}^{\text{glob}}_{\mathsf{G}, O_i})$

$$\mathcal{S}() := \mathsf{x}_v \leftarrow \mathcal{S}_1(); \quad \mathsf{x}_{res} \leftarrow \text{call } \mathsf{O}_i\ (\mathsf{x}_v)$$

Implementation in Squirrel

Invariant synthesis



Proof-search heuristic

- Goal oriented
- Constraints oriented
- Greedy on oracle calls

Incorporate in existing proof tool

Squirrel

- interactive proof assistant for protocol verification
- provides tactics for generic and cryptographic-specific reasoning

$\Rightarrow$ new tactic based on bideduction

And now ?

- Improve the proof-search heuristic (ongoing).
- "Stress test" on larger protocol (ongoing).
- Apply to other frameworks

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